| | |
|---|---|
| United States Patent Application Publication | 20250283996 |
| Kind Code | A1 |
| Publication Date | September 11, 2025 |
| Inventor(s) | Frischen; Andreas et al. |


## Method for Operating a Measuring Device


### Abstract

A computer-implemented method for operating a measuring device, in particular a wall diagnostic device includes (i) receiving radar data of a radar sensor unit of the measuring device, (ii) performing wall diagnostics by performing an analysis of the radar data and providing diagnostic results via a diagnostic module of the measuring device, (iii) providing, by way of the diagnostic module, the diagnostic results to a display unit of the measuring device, and (iv) displaying the diagnostic results in the display unit.



**Inventors:** **Frischen; Andreas (Leonberg, DE), Koelsch; Andreas (Peiting, DE), Giliard; Anke (Stuttgart, DE), Sgarz; Heiko (Leonberg, DE), Steuler; Jan Linus (Woelfersheim, DE), Borowski; Judith (Ho Chi Minh City, VN), Fiederer; Lukas (Balingen, DE), Troeltzsch; Uwe (Roesrath, DE)**

**Applicant:** **Robert Bosch GmbH** (Stuttgart, DE)



**Family ID:** **1000008528762**

**Appl. No.:** **19/071101**

**Filed:** **March 05, 2025**

### Foreign Application Priority Data

| | | |
|---|---|---|
| DE | 10 2024 202 231.4 | Mar. 11, 2024 |

### Publication Classification

**Int. Cl.:** **G01S13/88** (20060101); **G01S7/40** (20060101); **G01S7/41** (20060101); **G01S13/08** (20060101); **G01S13/86** (20060101)

**U.S. Cl.:**

CPC **G01S13/888** (20130101); **G01S7/4004** (20130101); **G01S7/41** (20130101); **G01S13/08** (20130101); **G01S13/86** (20130101);

## Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 202 231.4, filed on Mar. 11, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.
[0002] The present disclosure relates to a method for operating a measuring device, in particular a wall diagnostic device.
BACKGROUND
[0003] Diagnostic devices for performing wall diagnostics and detecting objects formed in the walls are known from prior art.
[0004] A task of the present disclosure to provide an improved method for operating a measuring device, in particular a wall diagnostic device.
[0005] The task is solved by the method set forth below. Advantageous embodiments are subject-matter also set forth below.
SUMMARY
[0006] According to one aspect, a computer-implemented method of operating a measuring device, in particular a wall diagnostic device is provided, wherein the method comprises: [0007] receiving radar data from a radar sensor unit of the measuring device, wherein the radar data depicts a wall on which diagnostics are to be performed; [0008] performing wall diagnostics by performing an analysis of the radar data and providing diagnostic results via a diagnostic module of the measuring device, wherein the wall diagnostics comprise: [0009] performing a wall type classification and determining a wall type of the wall by way of the diagnostic module, wherein the diagnostic results comprise at least the wall type of the wall; [0010] providing the diagnostic results via the diagnostic module to a display unit of the measuring device; and [0011] displaying the diagnostic results in the display unit.
[0012] The technical advantage can thereby be achieved that an improved method for operating a measuring device, in particular a wall diagnostic device, can be provided. For this purpose, radar data of a sensor unit of the measuring device is first received. The radar data depicts the wall on which diagnostics are to be performed. Furthermore, wall diagnostics are performed by a diagnostic module based on the radar data, and diagnostic results are provided. The wall diagnostics comprise at least determining a wall type of the wall on which diagnostics are to be performed, wherein the determined wall type is displayed as the diagnostic result in a display unit of the measuring device. With the present method, an automatic determination of a wall type of the wall on which diagnostics are to be performed may be provided based solely on radar data of a radar sensor unit. By automatically determining the wall type, a manual selection of the wall type of the wall to be examined by a user of the measuring device can be avoided. The wall type determined in the automatic wall type classification or automatic determination of the wall type can be further used in additional steps of the wall diagnostics, for example as background correction for an executed object recognition. This may further improve the quality of the wall diagnostics.
[0013] According to one embodiment, the wall diagnostics further comprise: [0014] determining uncertainty values for the diagnostic results by way of the diagnostic module, wherein the uncertainty values for the diagnostic results describe probability values that the diagnostic results match an actual state of the wall on which diagnostics are performed, and comprise at least one probability value for the wall type of the wall; and wherein the method further comprises: [0015] providing the uncertainty values along with the diagnostic results to a display unit using the

diagnostic module; and [0016] displaying the uncertainty values along with the diagnostic results in the display unit.

[0017] The technical advantage can thereby be achieved that additional information from the wall diagnostics can be provided to the user of the measuring device by determining uncertainty values of the diagnostic results, in particular the wall type determined in the wall diagnostics, and by providing and displaying the uncertainty values together with the diagnostic results on the display unit. The user can thus assess the diagnostic results based on the uncertainty values and can thus better adapt the further procedure when working on the wall to the executed wall diagnostics. If diagnostic results with a high uncertainty value are displayed, the user can assume that there is a high probability that the determined diagnostic result corresponds to an actual state of the wall to be examined. With low uncertainty values, the user can assume, on other hand, that the determined diagnostic results only reflect the actual state of the wall with a low probability. Taking into account the uncertainty values, the user can adjust the further procedure or the further work on the wall to the diagnostic results of the wall diagnostics. According to the disclosure, the uncertainty values describe probabilities for the accuracy of the provided diagnostic results.

[0018] According to one embodiment, a plurality of possible wall types of the wall will be determined as stand-alone diagnostic results in the wall type classification, wherein a probability value is determined for each of the plurality of wall types of the wall, and wherein the plurality of wall types of the wall are displayed as diagnostic results and the plurality of probability values of the different wall types are displayed as corresponding uncertainty values of the diagnostic results in the display unit.

[0019] The technical advantage can thereby be achieved that improved quality of the wall diagnostics is facilitated. For this purpose, a plurality of possible wall types of the wall are determined as stand-alone diagnostic results in the wall type classification. The respective different wall types are each provided with uncertainty values, which indicate a probability value that the respective wall type corresponds to the actual wall type of the wall to be examined. The user can evaluate the plurality of wall types displayed and decide which of the indicated wall types corresponds to the actual wall type of the wall to be examined.

[0020] According to one embodiment, the method further comprises: [0021] providing a first selection function, wherein at least one of the displayed wall types is selectable by performing the first selection function by a user of the measuring device.

[0022] The technical advantage can thereby be achieved that the selection function enables the user to select at least one of the plurality of displayed wall types as the actual wall type of the wall to be examined. The user thus has the option of incorporating their own knowledge of the wall to be examined into the wall diagnostics. Further improvement of the wall diagnostics can thereby be achieved.

[0023] According to one embodiment, the method further comprises: [0024] providing a second selection function, wherein, by the user of the measuring device performing the second selection function, automatic determination of the wall type during wall diagnostics and/or display of the wall type in the display unit can be deactivated, and a wall type is manually selectable by the user.

[0025] The technical advantage can thereby be achieved that, by way of the second selection function, the user can deactivate the automatic determination of the wall type and can manually select a wall type. The user can thus manually enter the actual wall type of the wall to be examined based on knowledge of the actual wall type, in order to further improve the wall diagnostics, especially in the event that the automatic wall type determination did not determine the actual wall type and thus provides an insufficient result. The wall type entered by the user can be used for further steps of wall diagnostics. This may further improve the quality of the wall diagnostics.

[0026] According to one embodiment, the method further comprises: [0027] displaying a plurality of possible wall types in the second selection function, wherein the user can select at least one of the displayed possible wall types in the second selection function.

[0028] The technical advantage can thereby be achieved that the wall diagnostics can be further improved. For this purpose, the user can select at least one of the multiple wall types indicated. This may improve matching the automatically determined wall type to the actual wall type present.
[0029] According to one embodiment, the wall diagnostics further comprise: [0030] performing an object recognition of an object located in the wall using the diagnostic module, wherein the object recognition comprises an object detection and an object classification, and wherein the diagnostic results comprise at least one object position in the wall and/or one object type of the object; and/or [0031] performing an object depth determination and determining an object depth of the object in the wall using the diagnostic module, wherein the object depth is defined as a distance of the object to a surface of the wall; and/or [0032] performing, by way of the diagnostic module, an object extension determination and determining an object extension of the object along a predefined direction, wherein the diagnostic results further comprise the object position in the wall and/or the object type and/or the object depth and/or the object extension of the object, and wherein the uncertainty values further comprise at least one probability value of the object position and/or a probability value of the object type and/or a probability value of the object depth and/or a probability value of the object extension of the object.
[0033] The technical advantage can thereby be achieved that the wall diagnostics can be further improved. For this purpose, in addition to determining the wall type, an object recognition including an object detection with a determination of an object position and an object classification with a determination of an object type is performed. Furthermore, an object depth and an object extension of an object located in the wall may be determined and provided as corresponding diagnostic results and displayed in the display unit.
[0034] According to one embodiment, the method further comprises: [0035] receiving selection commands from the user, wherein a wall type is selected by the user in the first or second selection function by the selection commands; [0036] determining, by way of the diagnostic module, the probability value of the selected wall type based on the radar data; and [0037] displaying the wall type for which the greatest probability value was determined based on the radar data if the probability value of the wall type selected by the user falls below a predetermined threshold value; and/or [0038] considering the wall type selected by the user in the second selection function for object recognition and/or object depth determination and/or object extension determination if the probability value of the selected wall type reaches or exceeds the predefined limit value; and/or [0039] considering, by way of the diagnostic module, the automatically determined wall type based on the radar data with the greatest probability value for object recognition and/or object depth determination and/or object extension determination.
[0040] The technical advantage can thereby be achieved that improved wall diagnostics are facilitated. If the user activates the second selection function and deactivates the automatic determination of the wall type and manually selects a wall type, the wall type manually entered by the user is only used for further wall diagnostics if the wall type entered has a probability value that reaches or exceeds a predefined threshold value. Otherwise, a wall type determined by the automatic wall type determination is used for the further wall diagnostics. This can prevent the subsequent wall diagnostics from being negatively affected by incorrect manual input of the wall type by the user.
[0041] According to one embodiment, the measuring device further comprises at least one of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor for providing additional sensor data, wherein the diagnostic module is configured to perform the wall diagnostics by taking into account the additional sensor data.
[0042] The technical advantage can thereby be achieved that, through further sensor data, additional sensors, each of which is configured to detect different physical variables, can provide additional information in addition to the radar data of the radar sensor unit, for incorporation into

the wall diagnostics. This additional information, which is preferably complementary to the information of the radar data of the radar sensor unit, allows further precise specification of the wall diagnostics or the object recognition.
[0043] According to one embodiment, the diagnostic module comprises at least one correspondingly trained artificial intelligence configured to perform an object recognition and/or wall classification and/or object depth determination based on the radar data and/or the additional sensor data
[0044] The technical advantage can thereby be achieved that, because the diagnostic module is configured as a correspondingly trained artificial intelligence structure, which is trained based on the radar data, or, if applicable, taking into account the information of the additional sensors, to perform an object recognition and/or a wall classification and/or an object depth determination and/or an object extension determination, a reliable and powerful diagnostic module can be provided. By using the artificial intelligence technology, precise wall diagnostics can be provided.
[0045] According to one embodiment, the object classes of the object type of the object comprise: metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, and/or wherein the wall type classes of the wall comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, underfloor heating, wall heating.
[0046] The technical advantage may thereby be achieved that objects and walls of different types may be detected and classified.
[0047] According to one aspect, a method for training an artificial intelligence structure of a wall diagnostic measuring device is provided, comprising: [0048] providing a training dataset for training an artificial intelligence, wherein the training dataset comprises a wall and radar data depicting an object formed in the wall as well as the feedback information provided according to the method for operating a measuring device; [0049] training the artificial intelligence based on the training dataset and taking into account the feedback information to perform object recognition of an object formed in a wall, wherein the object recognition comprises at least one object detection and one object classification.
[0050] The technical advantage can thereby be achieved that improved training of the artificial intelligence of the wall diagnostic device, in particular post-training, is facilitated, by way of which feedback is taken into account by the user of the wall diagnostic devices.
[0051] According to one aspect, a computing unit is provided, which is configured to perform the method for operating a measuring device according to one of the preceding embodiments.
[0052] According to one aspect, a computer program product is provided comprising instructions that, when the program is executed by a data processing unit, cause the data processing unit to perform the method for operating a measuring device according to one embodiment.

---

## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0053] Embodiments of the disclosure are described with reference to the following figures. The figures show:
[0054] FIG. **1** a schematic representation of a measuring device according to one embodiment;
[0055] FIG. **2** a further schematic representation of the measuring device according to a further embodiment;
[0056] FIG. **3** a further schematic representation of the measuring device according to a further embodiment;
[0057] FIG. **4** a schematic representation of a measurement of the measuring device according to

one embodiment,
[0058] FIG. **5** a further schematic representation of the measuring device according to a further embodiment;
[0059] FIG. **6** a schematic representation of a system for operating a measuring device according to one embodiment,
[0060] FIG. **7** a flowchart of a method for operating a measuring device according to one embodiment,
[0061] FIG. **8** another flowchart of the method for operating a measuring device according to a further embodiment,
[0062] FIG. **9** another flowchart of the method for operating a measuring device according to a further embodiment,
[0063] FIG. **10** another flowchart of the method for operating a measuring device according to a further embodiment, and
[0064] FIG. **11** a schematic representation of a computer program product.
DETAILED DESCRIPTION
[0065] FIG. **1** shows a schematic representation of a measuring device **100** according to one embodiment.
[0066] The present disclosure relates to a measuring device, in particular a wall diagnostic device for examining walls **105** to be worked on. Wall diagnostic devices are known from prior art that are used to detect objects located in walls. Such devices allow a user to examine walls to be worked on to search for objects located in the walls, in order to be able to perform planned work, for example drilling in walls, such that damage to the objects located in the walls can be avoided.
[0067] In the embodiment shown, the measuring device **100** comprises a housing **150** having a handle **152** for grasping of the measuring device **100** by a user, a display unit **111** for displaying diagnostic results **109** of the wall diagnostics, and controls **154** for switching the measuring device **100** between various operating modes.
[0068] According to the disclosure, the measuring device **100** comprises at least one radar sensor unit **101**. By way of the radar sensor unit **101**, radar signals may be transmitted towards the wall **105** to be examined, and radar signals reflected by the wall **105** may be received.
[0069] For example, the radar sensor unit **101** may be configured as a narrow band radar detector device in the 2.4 GHz to 2.4835 GHz frequency range or as an ultra-wide band radar detector device in the 1.8 GHz to 5.8 GHz frequency range.
[0070] The measuring device **100** further comprises a diagnostic module **107** executable on a computing unit **151** of the measuring device **100** to perform the wall diagnostics. The diagnostic module **107** is configured to perform corresponding diagnostics on the wall to be examined based on the radar data **103** of the radar sensor unit **101**. The radar data **103** of the radar sensor unit **101** thereby depicts the wall **105** to be examined and, if applicable, objects **113** located within the wall **105**.
[0071] The wall diagnostics carried out by the diagnostic module **107** comprise at least performing object recognition. The object recognition comprises an object detection and an object classification of the object **113** located in the wall **105**. The object detection comprises at least the determination of an object position **115**. The object position describes the positioning of the object located in the wall **105** with respect to a reference system defined by the measuring device **100**. The object classification of the detected object **113** comprises at least determining an object type **117** of the detected object **113**.
[0072] The diagnostic results of the wall diagnostics determined in this way, i.e. at least the determined object position **115** and/or the determined object type **117** of the object **113** located in the wall **105**, are subsequently presented in a display unit **111** of the measuring device **100** to a user of the measuring device **100**. For example, the display unit **111** may be configured as a corresponding display, and the diagnostic results **109** may be visually displayed. Additionally, the

display of the diagnostic results **109** may be supported via audible and/or haptic signals. For example, the haptic signals may be realized via corresponding vibration signals.
[0073] The object **113** can be shown in the display, for example, by way of a corresponding icon. The object **113** can be shown in the corresponding object position **115** in the display. The object extension **121** may be visualized by a corresponding size of the displayed icon. The particular object type **117** of the object **113** may be visualized with a corresponding term or color highlighting of the icon, or by a specific shape of the icon representing the object **113**.
[0074] Alternatively, the wall diagnostics may additionally comprise determining a wall type **123** in the form of a wall type classification of the wall **105** to be examined. The wall type **123** describes the respective type of the wall **105** to be examined. For example, the wall type may be associated with corresponding wall type classes, which may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or wall bricks, underfloor heating, wall heating or similar wall types found in buildings.
[0075] According to one embodiment, the diagnostic module **107** is further configured to determine an object depth **119** of the object **113** within the wall **105** based on the radar data **103**. The object depth **119** is defined by a distance of the object formed in the wall **105** to a surface of the wall **105**. The distance may be defined on the object side, for example with respect to an object surface or with respect to an object center point. The distance to the surface of the wall **105** describes a shortest distance defined by a direction perpendicular to the surface of the wall **105**.
[0076] According to one embodiment, the diagnostic module **107** is further configured to determine an object extension **121** of the object **113** in at least one predefined direction based on the radar data **103**. The object extension **121** of the object **113** describes a spatial extension of the object **113** in at least one spatial direction, preferably in two spatial directions, particularly preferably in three spatial directions. The object **113** may be described here as a one-dimensional, two-dimensional, or three-dimensional object **113**.
[0077] In conventional use, the measuring device **100** is placed on the surface of the wall **105** to be examined. Radar signals are transmitted towards the wall **105**, and radar signals reflected from the wall **105** or the objects **113** located behind it are received, via the radar sensor unit **101**. This radar data **103** of the radar sensor unit **101** is used by the diagnostic module **107** to perform the wall diagnostics described above and to determine corresponding diagnostic results **109**.
[0078] For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** of the object **113** located in the wall **105**. Alternatively or additionally, the diagnostic results **109** may comprise the wall type **123** of the wall **105** and/or the object depth **119** and/or the object extension **121** of the object **113**.
[0079] The diagnostic results **109** configured in this manner may subsequently be displayed to a user of the measuring device **100** in a display unit **111** of the measuring device **100**. The display unit **111** can be configured as a corresponding display, for example. The diagnostic results **109** may be displayed in graphical or textual form in the display unit **111**.
[0080] According to one embodiment, the measuring device **100** further comprises a motion detection unit **141**. The motion detection unit **141** may be used to detect movement of the measuring device **100** relative to the wall **105**. The motion detection unit **141** may comprise, for example, at least one roller element for this purpose. When the roller element is placed on the wall surface of the wall **105**, movement of the measuring device **100** relative to the wall **105** can be detected when the measuring device **100** moves along a direction of movement **153** by rolling the roller element. Alternatively, the motion detection unit **141** may have a different configuration by which a relative movement of the measuring device **100** relative to the wall **105** can be detected.
[0081] By moving the measuring device **100** relative to the wall **105**, radar data **103** of the radar sensor unit **101** may be captured for a plurality of different positions of the measuring device **100** relative to the wall **105**. This allows a larger spatial area of the wall **105** to be examined than the effective range of the radar sensor unit **101**. This allows for objects **113** to be captured that have a

greater spatial extent than the effective range of the radar sensor unit **101**.
[0082] While the measuring device **100** moves along the direction of movement **153**, radar data **103** of the radar sensor unit **101** may be captured continuously. The wall diagnostics may be evaluated by the diagnostic module **107** based on this radar data **103** while the measuring device **100** is moving along the direction of movement **153**. This allows for accelerated wall diagnostics, taking into account the positioning of the measuring device **100** relative to the wall **105**.
[0083] According to its embodiment, the diagnostic module **107** is configured as a correspondingly trained artificial intelligence **125**. The artificial intelligence **125** is trained to perform the above-mentioned wall diagnostics based on the radar data **103** of the radar sensor unit **101** and to determine at least the object position **115** and the object type **117** of an object **113** located in the wall **105**. The object classification or determination of the object type **117**, respectively, comprises assigning the detected object **113** to predefined object classes.
[0084] The object classes may comprise: Metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, or other elements commonly installed in building walls.
[0085] Furthermore, the artificial intelligence **125** may be trained to determine the wall type **123** of the wall **105** to be examined at least based on the radar data **103** of the radar sensor unit **101**. Possible wall types **123** may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or individual bricks of the brick wall, underfloor heating, wall heating or other similar wall types commonly installed in buildings.
[0086] According to one embodiment, in addition to the radar sensor unit **101**, the measuring device **100** may comprise further additional sensors by way of which additional physical variables are detectable. For example, the measuring device **100** may comprise an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor or other sensors commonly used in wall diagnostic devices.
[0087] The diagnostic module **107**, in particular the corresponding trained artificial intelligence **125**, can be configured to perform wall diagnostics based on the radar data **103** of the radar sensor unit **101** and taking into account the additional sensor information of the further sensors described above. The additional information of the additional sensors mentioned above can particularly be used for object recognition of the objects **113** located in the walls **105**. The additional sensor information may possibly provide improved detection of the objects **113** and may possibly provide improved classification of the objects **113**.
[0088] For example, the material of the objects **113**, for example as a metallic or non-metallic material, can be particularly improved and classified by using the additional sensor information.
[0089] FIG. **2** shows another schematic representation of the measuring device **100** according to a further embodiment.
[0090] In the embodiment shown, the measuring device **100** comprises a pre-processing module **127** in addition to the diagnostic module **107**. For wall diagnostics, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. Pre-processing of the receiving radar data **103** is performed via the pre-processing module **127**. For example, via pre-processing by the pre-processing module **127**, the radar data may be brought into a corresponding data structure required for wall diagnostics by the diagnostic module **107**.
[0091] As described above, during wall diagnostics, the diagnostic module **107** generates the diagnostic results **109** described above. For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** and/or object depth **119** and/or object extension **121** of an object **113** formed in the wall **105** to be examined and/or the wall type **123** of the wall **105** to be examined. The correspondingly generated diagnostic results **109** may subsequently be displayed in the display unit **111** of the measuring device **100**.
[0092] According to one embodiment, in addition to the radar data **103** of the radar sensor unit **101**,

the additional sensor information of the additional sensors described above may be considered during the wall diagnostics of diagnostic module **107**. A corresponding pre-processing of the additional sensor information may be performed accordingly by the pre-processing module **127**.
[0093] In the embodiment shown, the diagnostic module **107** comprises a wall type classification module **129** and an object recognition module **131**. The pre-processing module **127** comprises a first pre-processing module **135** and a second pre-processing module **137**. The first pre-processing module **135** comprises an S matrix reduction **155**. The second pre-processing module **137** comprises a background correction **157**, an inverse Fast Fourier transformation **159**, and a focusing and migration **161**. During the pre-processing of the radar data **103** by the pre-processing module **127**, the radar data **103** is first pre-processed by the first pre-processing module **135** and the S-matrix reduction **155** contained therein.
[0094] When doing so, the first pre-processing module **135** generates input data **133** based on the radar data **103**. The input data **133** serves as input data for the wall type classification module **129**. The wall type classification module **129** performs a wall type classification of the wall **105** to be examined based on the input data **133** and generates wall type information **139**. The wall type information **139** contains the wall type **123** of the wall **105** to be examined, as determined in the wall type classification.
[0095] Subsequently, the second pre-processing module **137** performs pre-processing based on the radar data **103** and the wall type information **139**. A background correction **157** of the radar data **103** is performed during this, taking into account the wall type **123** determined in the wall type information **139**. Depending on the wall type **123** of the wall **105** to be examined, different effects can occur on the radar data **103**.
[0096] These effects, which are primarily based on the respective wall type **123** and can affect object recognition, can be corrected by the background correction **157**. After the background correction has been performed, further pre-processing can be carried out by performing the inverse Fast Fourier transformation **159** or focusing and migration **161**, respectively, and input data **133** can be created for the object recognition module **131** once again. Based on the input data **133** provided by the second pre-processing module **137**, the object recognition module **133** performs object recognition of the object **113** located in the wall **105** to be examined and determines at least the object position **115** and the object type **117** of the respective object **113**. Additionally, the object depth **119** and the object extension **121** may be determined by the object recognition module **131**.
[0097] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.
[0098] The pre-processing is optional. Depending on the algorithm used for the diagnostic module **107**, completely unprocessed [0099] radar echoes of different frequencies may be used as radar data **103** and as input data for the diagnostic module **107**. Alternatively, radar data **103** processed via multiple steps may be used. The pre-processing steps comprise, for example, the transformation of the [0100] signals from the frequency domain to the time or distance domain, background deduction, noise removal, and normalization of the signals. For radar data **103** which is available in the form of complex numbers, only the absolute value can be processed. Alternatively or additionally, the phase information may be considered.
[0101] FIG. **3** shows a further schematic representation of the measuring device **100** according to a further embodiment.
[0102] In the embodiment shown, the diagnostic module **107** comprises a plurality of parallel processing paths **102**. Each processing path **102** includes a pre-processing module **127**, the diagnostic module **107**, comprising the wall type classification module **129** and/or the object recognition module **131**, for example, according to the embodiment in FIG. **2**, and a post-processing module **163**.
[0103] In FIG. **3**, primarily the radar data **103** is shown as input data for the wall diagnostics. In

addition to the radar data shown, however, the additional information from the additional sensors may also serve as input data for the wall diagnostics. The different information from the different types of sensors in the different parallel processing paths **102** can be processed and the corresponding wall diagnostics performed separately on the different sensor information. Upon completion of the wall diagnostics, a summary module may be used to assemble a summary of the individual partial analysis results for the diagnostic results **109** of the wall diagnostics.

[0104] Alternatively or additionally, different sub-aspects of wall diagnostics may be performed by the different processing paths **102** based on the same sensor information.

[0105] For example, the individual processing paths **102** may process different radar data **103** captured during movement of the measuring device **100** relative to the wall **105** for different positions of the measuring device **100** relative to the wall **105**. The radar data **103**, thus representing different regions of the wall **105** and captured sequentially in time as the measuring device **100** moves relative to the wall **105**, may then be processed in the different processing paths **102** by the modules shown.

[0106] The various processing paths perform stand-alone wall diagnostics including at least determining the object position **115** and/or the object type **117** of the object **113** located in the wall **105**.

[0107] By way of the summary module **165**, the partial results of the stand-alone wall diagnostics of the different regions of the wall **105** provided in the individual processing paths **102** can be summarized to a contiguous diagnostic result **109**. The contiguous diagnostic result describes the wall diagnostics of a contiguous spatial area that was scanned during movement of the measuring device **100** relative to the wall **105** and depicted by the corresponding captured radar data **103**. The parallel processing of the radar data **103** or additional sensor information **104** of the additional sensor elements in the different processing paths **102** thus enables accelerated wall diagnostics.

[0108] Alternatively, various wall diagnostic functions may also be performed in the different processing paths **102**. For example, in one processing path **102**, the wall type classification and the determination of the wall type **123** of the wall **105** to be examined can be performed. In another processing path **102**, object recognition of the object **113** located in the wall can be performed. The object detection can be performed with the determination of the object position **115** and the object classification can be performed with the determination of the object type **113** in one processing path **102**.

[0109] Alternatively, the object detection and object classification may then also be performed in two separate processing paths **102**. In further processing paths **102**, the object depth determination, i.e., the determination of the object depth **119** and/or the determination of the object extension **121** may each be carried out. In the summary module **165**, the various partial results of the wall diagnostics may be summarized into corresponding diagnostic results **109**.

[0110] The diagnostic module **107** can be divided into different artificial intelligence structures **125**, as already shown in the embodiment in FIG. **2**. For example, the diagnostic module **107** may comprise a wall type classification module **129** and an object recognition module **131**. The object recognition module may in turn be divided into an object detection module and an object classification module. The diagnostic module **107** may further comprise an object depth determination module and object extension module, respectively configured to determine the object depth **119** and the object extension **121**.

[0111] The respective modules may each be configured as stand-alone artificial intelligence structures **125**, for example, neural networks. Alternatively, the various modules may form portions of an overall artificial neural network that are connected to an overall neural network according to structures known from prior art.

[0112] FIG. **4** shows a schematic illustration of a measurement by the measuring device **100** according to one embodiment.

[0113] For pre-processing, the radar data **103** or the additional sensor information **104** of the

remaining sensors may be normalized, in particular to numerically stabilize the subsequent steps performed by the diagnostic module **107** during the wall diagnostics. For example, an amplitude and/or offset compensation may be performed for this purpose. Moreover, to reduce interference, filtering of the radar data **103** may be carried out, and to reduce the data rate, the corresponding sensor data may be sampled. Additionally, the radar data **103** or the additional sensor information **104** may be transformed into the respectively required frequency range or time range. Methods known from prior art can be used for this purpose.
[0114] Furthermore, the captured radar data **103** or additional sensor information **104** may be divided into temporal or spatial windows **167**. Temporal windows **167** may be generated by recording the radar data **103** or the additional sensor information or the pre-processed radar data **103** over a fixed time interval. Spatial windows **167**, on the other hand, may be generated from a mapping of the radar data **103** or additional sensor information **104** to positions of the measuring device **100** relative to the wall **105** along the direction of movement **153**.
[0115] Graph a) of FIG. **4** shows such a data matrix resulting from the steps described above. The data matrix of the window **167** shown in graph a) shows a plurality of sensor data, which may comprise, for example, radar data **103** or additional sensor information **104** of the further sensors plotted along a frequency channel axis **171** or along a space/time axis **169**, respectively.
[0116] A width of the temporal windows **167** may be selected such that different sampling rates of the sensors may be balanced and a new window **167** may be provided frequently enough so that a display of the diagnostic results **109** of the wall diagnostics in the display unit **111** may be shown without too great of a time delay while the measurement is being performed or shortly after the measurement by the measuring device **100** ends.
[0117] A rate of 2 to 20 windows per second of the data recording of the sensor data may be advantageous for this purpose. For spatial windows, the spatial sampling rates can be selected to achieve the desired local accuracy. Advantageously, 1 mm to 1 cm sampling rates can be used. This means that corresponding sensor data is captured every 1 mm to 1 cm of movement of the measuring device **100** along the direction of movement **153**.
[0118] A width of the spatial windows **167** may be selected such that contiguous information relating to an object **113** is contained in a window. Advantageously, a width of the respective spatial windows **167** can be 1 cm to 20 cm. This results in 4 to 100 measured values per window **167**. This allows for further efficient algorithmic processing of the correspondingly recorded radar data **103** or additional sensor information by the diagnostic module **107**.
[0119] A further temporal window **167** or spatial window **167** can be provided as soon as one or more sampling points are available.
[0120] The diagnostic module **107** may be oriented such that a matrix corresponding to the window size of the respective spatial or temporal window **167** may be included as input data, for example of each processing path **102** of the embodiment in FIG. **3** as well. According to the embodiment of FIG. **2**, the corresponding input data may comprise the respective pre-processed sensor data, i.e. radar data **103** and additional sensor information **104** of the additional sensors.
[0121] As stated above, the wall diagnostics may be performed by the diagnostic module **107** based on a correspondingly trained artificial intelligence. Alternatively, different processing paths may also be calculated by way of rule-based algorithms. In addition, within a processing path **102**, a combination of artificial intelligence and rule-based algorithm is possible in the form of a parallel operation or concatenation.
[0122] The diagnostic results **109** of the wall diagnostics may be expressed as numeric values, vectors, or matrices. Furthermore, for the object detection, the probability of detection, or for the wall type classification and/or the object classification, respectively, a probability of the specified object classes and/or wall type classifications can be indicated. The same may apply to the position and/or depth determination, for which corresponding probability values can also be indicated.
[0123] In addition to the radar data **103**, if the additional sensor information of the further sensor

types is processed in a processing path **102**, these may either be merged within the artificial intelligence **125** or combined by way of rule-based combinations.
[0124] During the post-processing of each processing path **102**, of the embodiment in FIG. **3**, multiple algorithm results based on multiple windows **167** may be summarized by the summary module **165**. This summary can in particular be realized by way of majority formation, sum formation or also by multiplication of successive probability values.
[0125] Furthermore, by clustering multiple results, for example, it is possible to detect which objects, of multiple detected objects located close to one another, are the same object so that they are not incorrectly detected multiple times.
[0126] Likewise, it is possible to multiplicatively apply a weighting function **177** when summarizing the results from multiple windows **167**. Advantageously, the diagnostic partial results **175** corresponding to corresponding data points in the space may be weighted with respect to positioning of the diagnostic partial results **175** relative to a center point of the respective window **167**. This is illustrated by way of example in graph b), in which the individual diagnostic partial results **175** are weighted according to the weighting function **177** shown with respect to the center point of the window **167** shown.
[0127] According to one embodiment, the results of one processing path **102** after post-processing **163** may influence the extension of another processing path **102**. Weighting parameters may be adjusted that may depend on the particular result from the processing path **102** for each window.
[0128] For example, the result of an object classification in which the object type **117** of an object located in the wall **105** is defined, can be used to increase the weight of a wall type classification, in which the wall type **123** of the respective wall **105** is determined, during the post-processing at locations without objects **113**, because the respective radar data **103** at these locations is less influenced by reflections of the objects **113**.
[0129] FIG. **5** shows a further schematic representation of the measuring device **100** according to a further embodiment.
[0130] Graphs a) and b) of FIG. **5** show two different alternatives of joint data processing of radar data **103** and additional sensor information **104** by the diagnostic module **107**.
[0131] Graph b) illustrates a joint processing of the radar data **103** and the additional sensor information **104** of the additional sensors by the diagnostic module **107**. For this purpose, the radar data **103** and the additional sensor information **104** are collectively used as input data of the diagnostic module **107** configured as an artificial intelligence, in particular as an artificial neural network. The diagnostic module **107** here comprises multiple convolutions **108** and multiple dense layers **106**. The radar data **103** and the additional sensor information **104** are processed jointly as input data via the convolutions **108** and dense layers **106**. The aforementioned diagnostic results **109** are created as output data of the diagnostic module **107** based on this.
[0132] On the other hand, in graph b) the radar data **103** and the additional sensor information **104** are used as stand-alone input data of the diagnostic module **107**. The diagnostic module **107** becomes multiple processing paths **102**. The processing paths **102** each comprise multiple convolutions **108** and at least one dense layer **106**. In the various processing paths **102**, wall diagnostics are performed separately by the diagnostic module **107** based on the radar data **103** and the additional sensor information **104**, respectively.
[0133] In an additional concatenation layer **148**, the partial results of the partial diagnostics of the different processing paths **102** are combined and fed to a final dense layer **106**. The output data of the diagnostic module **107** corresponds to the diagnostic results **109** described above.
[0134] The correspondingly configured diagnostic module **107** is configured to perform wall diagnostics as described above, including the features described above, based on the radar data **103** and the additional sensor information **104**.
[0135] In the embodiment shown, the diagnostic module **107** is configured as an artificial neural network, in particular as a convolutional network. Corresponding network architectures with

convolutions **108**, dense layers **106**, and concatenation layers **148** are known from prior art.
[0136] FIG. **6** shows a schematic representation of a system **600** for operating a measuring device **100** according to one embodiment.
[0137] In the embodiment shown, the system **600** comprises at least the measuring device **100** including the diagnostic module **107**.
[0138] According to the disclosure, at least one wall type classification based on the radar data **103** is performed by the diagnostic module **107** to carry out the wall diagnostics, and at least the wall type **123** of the wall **105** is determined. The respective wall type **123** is provided to the display unit **111** as a diagnostic result **109** and is displayed thereon.
[0139] In addition to the wall type **123**, the object position **115** and/or the object type **117** and/or the object depth **119** and/or the object extension **121** can also be determined as the corresponding diagnostic result **109** in the wall diagnostics.
[0140] In the embodiment shown, an uncertainty value **110** is further determined by the diagnostic module **107** for each determined diagnostic result **109**. The uncertainty value describes a probability value that the respective diagnostic result **109** reflects the actual state of the wall **105** to be examined. In the embodiment shown, the uncertainty value **110** thus defines the probability value that the indicated wall type **123** corresponds to the actual wall type **123** of the wall **105** to be examined.
[0141] In the embodiment shown, a plurality of possible wall types **123** are determined by the diagnostic module **107** during wall diagnostics. The determined wall types **123** are each shown as stand-alone diagnostic results **109** in the display unit **111**. Each possible wall type determined is provided with a corresponding uncertainty value **110**.
[0142] The diagnostic module **107** is configured to calculate the respective probability value for each of the determined possible wall types **123** based on the wall diagnostics performed.
[0143] According to the embodiment shown in graph a), the measuring device **100** further provides a first selection function **116** and a second selection function **118**. By way of the first selection function **116**, the user can select a diagnostic result **109** to be considered for further wall diagnostics among the plurality of diagnostic results **109**, for example among the plurality of wall types **123** shown. The unselected diagnostic results **109** are thus disregarded for further wall diagnostics. In the embodiment shown, the user can thus select one of the plurality of possible wall types **123** displayed for further wall diagnostics via the first selection function **116**.
[0144] The second selection function **118**, on the other hand, allows the user to deactivate the automatic wall type determination. The user may further manually input a wall type **123**, on which the further wall diagnostics are to be performed.
[0145] According to one embodiment, in the event that a wall type **123** was manually input by the user via the second selection function **118**, the diagnostic module **107** is configured to determine an uncertainty value **110** for this wall type **123** based on the wall diagnostics. Furthermore, based on the determined uncertainty value **110**, the diagnostic module **107** is configured to take into account the wall type **123** manually entered by the user for further wall diagnostics if the determined uncertainty value **110** reaches or exceeds a predefined threshold value. On the other hand, if the determined uncertainty value **110** of the wall type **123** selected by the user does not reach the predefined threshold value, this can be indicated to the user on the display unit **111**. In this case, for example, the user may be shown one of the wall types **123** automatically determined during wall diagnostics as a possible wall type. Alternatively, several of the determined wall types **123** can also be displayed to the user. For further wall diagnostics, one of the automatically determined wall types **123**, preferably the one with the highest uncertainty value **110**, can be considered.
[0146] Graph b) shows another embodiment for actuating the second selection function **118**. When deactivating the automatic wall type determination by actuating the second selection function **118**, a plurality of pre-saved wall types **123** can be displayed to the user as an alternative or in addition to the possibility of the user manually entering a wall type **123**. The user can thus select one of the

possible wall types **123** stored, for example, in a database **120**.
[0147] FIG. **7** shows a flowchart of a method **200** for operating a measuring device **100** according to one embodiment.
[0148] To operate the measuring device **100**, in a method step **201**, radar data **103** of the radar sensor unit **101** of the measuring device **100** is first received. The radar data **103** depicts the wall **105** on which diagnostics are to be performed.
[0149] In another method step **203**, wall diagnostics are performed by the diagnostic module **107** based on the radar data **103**, and diagnostic results **109** are provided.
[0150] For this purpose, in a method step **205**, a wall type classification is performed and a wall type **123** of the wall **105** is determined by the diagnostic module **107**.
[0151] In another method step **207**, the diagnostic results are provided to the display unit **111** by the diagnostic module **107**.
[0152] In a method step **209**, the diagnostic results **109** are displayed in the display unit **111**.
[0153] FIG. **8** shows another flowchart of the method **200** for operating a measuring device **100** according to a further embodiment.
[0154] The embodiment shown in FIG. **8** is based on the embodiment in FIG. **7** and comprises all the method steps described there.
[0155] In the embodiment shown, in a method step **211**, during wall diagnostics, uncertainty values **110** of the diagnostic results **109** are determined by the diagnostic module **107**. The uncertainty values **110** describe probability values that the diagnostic results **109** match the actual state of the wall **105** to be examined.
[0156] In method steps **207** and **209**, the uncertainty values determined in method step **211** with the corresponding diagnostic results **109** are provided to and displayed in the display unit **111**.
[0157] FIG. **9** shows another flowchart of the method **200** for operating a measuring device **100** according to a further embodiment.
[0158] The embodiment shown in FIG. **9** is based on the embodiment in FIG. **8** and comprises all the method steps described there.
[0159] In the embodiment shown, in a method step **219**, during the course of wall diagnostics, an object detection of the at least one object **113** disposed in the wall **105** is performed with the object detection including the determination of the object position **115** and the object classification including the determination of the object type **117**.
[0160] In a further method step **221**, the object depth determination is performed and the object depth **119** of the object **113** is determined.
[0161] In a method step **223**, the object extension determination is performed and the object extension **221** of the object **113** is determined.
[0162] The object position **115** and/or the object type **117** and/or the object depth **119** and/or the object extension **121** are provided as corresponding diagnostic results **109** of the display unit **111** and displayed thereon. For the object position **115** and/or the object type **117** and/or the object depth **119** and/or the object extension **121**, corresponding uncertainty values **110** are calculated and shown on the display unit **111**.
[0163] FIG. **10** shows another flowchart of the method **200** for operating a measuring device **100** according to a further embodiment.
[0164] The embodiment shown in FIG. **10** is based on the embodiment in FIG. **8** and comprises all the method steps described there.
[0165] According to the embodiment shown, a plurality of possible alternative results is provided by the diagnostic module **107** during the wall diagnostics as a diagnostic result **109**. For example, multiple different possible wall types **123** are determined and may be displayed on the display unit **111**. The plurality of wall types **123** are provided with corresponding uncertainty values **110**. The same applies to the object position **115**, the object type **117**, the object depth **119** and the object extension **121**, for which several possible alternative values including the uncertainty values **110**

can also be determined.
[0166] In a further method step **213**, a first selection function **116** is provided. By activating the first selection function **116**, at least one of the plurality of possible wall types **123** indicated may be selected.
[0167] Furthermore, in a method step **215**, a second selection function **118** is provided. By activating the second selection function **118**, the automatic determination of the wall type **123** may be deactivated. In addition, a wall type **123** is manually selectable by the user.
[0168] In a further method step **217**, the plurality of possible wall types **123** is displayed in the second selection function **118**. The user may select at least one of the displayed wall types **123** by activating the second selection function **118**.
[0169] In a further method step **225**, selection commands of the user are received by the measuring device **100**, wherein a wall type **123** is selected by the selection commands.
[0170] In a further method step **227**, a corresponding uncertainty value in the form of a probability value is ascertained or determined for the selected wall type **123**.
[0171] In a further method step **229**, the wall type **123** for which the greater probability value was determined based on the radar data **103** is displayed in the display unit if the probability value of the wall type **123** selected by the user falls below a predetermined threshold value.
[0172] In a further method step **235**, the probability values of the determined wall types **123** are checked.
[0173] In a further method step **231**, the wall type **123** selected by the user in the second selection function **118** is considered for further wall diagnostics if the probability value of the selected wall type **123** reaches or exceeds the predefined limit value.
[0174] If the probability value of the selected wall type does not reach the predetermined limit value, then, in a further method step **233**, a wall type **123** determined during the automatic determination of the wall type **123** with the greatest uncertainty value **111** is considered for the further wall diagnostics.
[0175] FIG. **11** shows a schematic representation of a computer program product **500** comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method **200** for operating a measuring device **100**.
[0176] In the embodiment shown, the computer program product **500** is stored on a storage medium **501**. The storage medium **501** can, in this case, be any desired storage medium known from prior art.

## Claims

**1**. A computer-implemented method of operating a measuring device, comprising: receiving radar data of a radar sensor unit of the measuring device, wherein the radar data depicts a wall on which diagnostics are to be performed; and performing wall diagnostics by performing an analysis of the radar data and providing diagnostic results by a diagnostic module of the measuring device, wherein the performing wall diagnostics comprises: performing a wall type classification and determining a wall type of the wall by way of the diagnostic module, wherein the diagnostic results comprise at least the wall type of the wall; providing the diagnostic results by the diagnostic module to a display unit of the measuring device; and displaying the diagnostic results in the display unit.
**2**. The method according to claim 1, wherein: the performing wall diagnostics further comprises determining uncertainty values of the diagnostic results by the diagnostic module, the uncertainty values of the diagnostic results describe probability values that the diagnostic results match an actual state of the wall on which diagnostics are performed, and comprise at least one probability value of the wall type of the wall, and wherein the method further comprises (i) providing the uncertainty values together with the diagnostic results by the diagnostic module to the display unit,

and (ii) displaying the uncertainty values along with the diagnostic results in the display unit.
**3**. The method according to claim 1, wherein: a plurality of possible wall types of the wall will be determined as stand-alone diagnostic results in the wall type classification, a probability value is determined for each of the plurality of wall types of the wall, and the plurality of wall types of the wall are displayed as diagnostic results and the plurality of probability values of the different wall types are displayed as corresponding uncertainty values of the diagnostic results in the display unit.
**4**. The method according to claim 3, further comprising: providing a first selection function, wherein at least one of the displayed wall types is selectable by performing the first selection function by a user of the measuring device.
**5**. The method according to claim 3, further comprising: providing a second selection function, wherein, by the user of the measuring device performing the second selection function, automatic determination of the wall type during wall diagnostics and/or display of the wall type in the display unit is deactivated, and a wall type is manually selectable by the user.
**6**. The method according to claim 5, further comprising: displaying a plurality of possible wall types in the second selection function, wherein the user is able select at least one of the displayed possible wall types in the second selection function.
**7**. The method according to claim 1, wherein the performing wall diagnostics further comprises: performing an object recognition of an object located in the wall using the diagnostic module, wherein the object recognition comprises an object detection and an object classification, and wherein the diagnostic results comprise at least one object position in the wall and/or one object type of the object; and/or performing an object depth determination and determining an object depth of the object in the wall using the diagnostic module, wherein the object depth is defined as a distance of the object to a surface of the wall; and/or performing, by way of the diagnostic module, an object extension determination and determining an object extension of the object along a predefined direction, wherein the diagnostic results further comprise the object position in the wall and/or the object type and/or the object depth and/or the object extension of the object, and wherein the uncertainty values further comprise at least one probability value of the object position and/or a probability value of the object type and/or a probability value of the object depth and/or a probability value of the object extension of the object.
**8**. The method according to claim 5, further comprising: receiving selection commands from the user, wherein a wall type is selected by the user in the first or second selection function by the selection commands; determining, by way of the diagnostic module, the probability value of the selected wall type based on the radar data; and displaying the wall type for which the greatest probability value was determined based on the radar data if the probability value of the wall type selected by the user falls below a predetermined threshold value; and/or considering the wall type selected by the user in the second selection function for object recognition and/or object depth determination and/or object extension determination if the probability value of the selected wall type reaches or exceeds the predefined limit value; and/or considering, by way of the diagnostic module, the automatically determined wall type based on the radar data with the greatest probability value for object recognition and/or object depth determination and/or object extension determination.
**9**. The method according to claim 1, wherein object classes of the object type of the object comprise: metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, non-water filled plastic pipe, and/or wherein the wall type classes of the wall type of the wall comprise: concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, underfloor heating, wall heating.
**10**. The method according to claim 1, wherein: the measuring device further comprises at least one of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor for providing additional sensor

data, and the diagnostic module is configured to perform the wall diagnostics by taking into account the additional sensor data.

**11**. The method according to claim 1, wherein the diagnostic module comprises at least one correspondingly trained artificial intelligence configured to perform object detection and/or a wall classification and/or an object depth determination and/or an object extension determination based on the radar data and/or the additional sensor data.

**12**. A computing unit configured to perform the method for operating a measuring device according to claim 1.

**13**. A computer program product comprising instructions which, when the program is executed by a data processing unit, cause the data processing unit to perform the steps of the method for operating a measuring device according to claim 1.

**14**. The method according to claim 1, wherein the measuring device is a wall diagnostic device.

**15**. The method according to claim 9, wherein: the water filled plastic pipe includes a fresh water pipe, and the non-water filled plastic pipe includes a waste water pipe.

---